Boston College, Department of Physics, Spring 2009
PH210 Introductory Physics Syllabus

Scheduled class meeting times:
Monday 9:00 AM, Wednesday 9:00 AM, Thursday 3:00 PM, Friday 9:00 AM
Room = Higgins 225

Required Text: *Physics, Vol. 1 & 2,* 5th ed*,* Halliday, Resnick, and Krane (Wiley & Sons, New York, 2002)

Instructor:
Fr. Cyril P. Opeil, S.J., Ph.D.
Assistant Professor
Office: Higgins 330K, 617-552-8450
Laboratory: Higgins 130, 617-552-3589
Office Hours: As posted at 330K

Office Hours: As posted at 330K:
Monday 11:00AM-NOON
Wednesday 11:00AM-NOON, 1:00-3:00PM
and by appointment.

To contact Fr. Opeil, S.J. outside of class the most efficient way is via email: opeil@bc.edu, please feel free to call at 2-8450 or 2-3589 but do not leave a phone message.

Graduate Assistants and Home Work Graders:
Shankar Kunwar (2-6025, Higgins 031, shankaku@bc.edu) and Zhensong Ren (2-3436, Higgins 160, renzd@bc.edu) are the teaching assistants assigned to this class. Their primary job is to grade homework assignments, provide assistance on homework problems upon request, and assist the instructor in correcting the exams.

Pre-requisites: PH203, PH209 and MT102, (Advance Placement AB or BC are highly recommended though not required). If you have no background in Physics or have never taken a course in physics, this is not the place to start.

Co-requisites: 2nd semester of introductory calculus (MT103 or equivalent). Students should also be enrolled in Physics Laboratory II - PH204.

Grading:	3 mid-term exams (15% each)	45 %
	1 final exam (comprehensive)	20 %
	homework	25 %
	class participation	10 %

Course Description: This course (4 credit) is designed as an introduction into the fundamentals of electrostatics, magnetism, electricity, basics of electrical circuits, electromagnetism, Maxwell's Equations, light, and optics. This course makes much more use of integral calculus and differentiation than during the first semester, and in general the math level will be higher.

Course Goals: As for the second semester, the goals of this course are: (1) to better understand the fundamental principles (forces and motion) governing, charged particles, electricity and magnetism (2) to formulate these principles in mathematical terms; and (3) to be able to use these mathematical expressions to predict the behavior of simple systems. This the first course of three in which the principles of electricity and magnetism are taught. See books by Griffiths (undergraduate) and Jackson (graduate) for details.

Attendance, Participation and Academic Honesty: Continuity and active participation are crucial to the success of this course. You are responsible for all information from each class lecture session whether you attend or not. Absences due to excused absence sports activities are to be discussed with the instructor prior to the planned absence.

Academic honesty is expected at all times in accordance with published Boston College policies. Discussion of the concepts and problem-solving techniques between students, and with the instructor and graders, is strongly encouraged. However, under **no circumstances is it acceptable to copy and turn in solutions taken from ANY source** – all work must be your own. Please review the Boston College Standards of Academic Integrity that are published in the university catalog and on the web at:

http://www.bc.edu/offices/stserv/academic/resources/policy/#integrity .

It is *your responsibility* to ensure that all work you submit is in accordance with university policies. If you have any questions, please consult with me. Violations will be reported to the Deans' Office and reviewed by the College's Committee on Academic Integrity. Sanctions for violating this policy include the possibility of failing this course.

Textbook Reading Assignments: Members of this course are responsible for reading particular pages in the textbook on a weekly basis. Textbook sections listed in the schedule are to be read **prior** to attending class. Any changes in the textbook reading assignments schedule will be announced in class.

Homework Exercises: Expect homework assignments each week, note schedule below. In order to receive FULL CREDIT for your problem solutions you must *fully explain* your solutions: briefly outline the logic of your approach to the solution in words, and write out all equations used to obtain your solution. For multiple choice questions, the answer requires more than a letter choice, you must include why that choice is correct. Please be neat: if we can't read it, we can't grade it properly. ***Students are strongly encouraged to discuss problems with the instructor*** – during office hours – they are posted. The course calendar/schedule below lists when assignments are due. ***Actually doing the problems is not the equivalent of copying the problems from another student's solution.***

Homework solutions are due from students at the beginning of class on the assigned day. Homework solutions will be emailed to students one hour after student solutions are turned in according to the schedule below. Homework solutions may be emailed to the instructor (Opeil@bc.edu) **prior** to this date and time if the student plans not to attend class for any reason. Homework assignments **will not be accepted after the beginning of class on the assigned dates**. For problem set details see schedule below. Corrections and emendations to the homework schedule may be made by the instructor as the course progresses.

Class participation: Students are expected to participate in class discussions and exercises, and will be given a grade based on that participation. Clearly you need to attend class to participate, and attendance in classes will be monitored and will count towards your grade.

Midterm and Final Exams: Three written midterms (50 minutes each) and a cumulative final exam (two hours) are to be given according to the Course Schedule listed below. *No one is allowed to take the final at any other time* without **written permission** of the Dean.

Disabilities: If you have a disability and will be requesting accommodations for this course, please register with Kathy Duggan [kathleen.duggan@bc.edu], Associate Director, Academic Support Services, The Connors Family Learning Center (learning disabilities and ADHD) or

Suzy Conway [suzy.conway.1@bc.edu], Assistant Dean for Students with Disabilities (all other disabilities). Advanced notice and appropriate documentation are required for accommodations.

Eating and Drinking: *Absolutely* no food or beverages are to be brought into or consumed in the lecture room. Not consuming of food and or beverages in the lecture hall is a matter of simple courtesy and civility.

Class Resources:
A variety of materials will be on 2-hour reserve for PH210 in O'Neill Library.

Extra Help:
Please see the instructor if you have any questions or concerns about this course or course material. See graduate student graders or teaching assistants assigned to this course for additional help. Also, there are additional resources. You may drop in on afternoons without an appointment to the Department of Physics' "**Undergraduate Resource Room**" (URR), located in Higgins 223, across the hall from our classroom. On the 2nd floor of O'Neill Library, the **Connors Family Learning Center** has tutors available for introductory physics. Sessions are free of charge, but you must make your appointments in advance.

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Schedule- Spring 2009 PH210: MWF=9, Th=3

W-14jan09	Review Course Syllabus, 14.1-14.3	
Th-15jan09	14.4-14.6	
F-16jan09	14.7-problems	
M-19jan09	Martin Luther King-no classes	
W-21jan09	25.1-25.4	PS0 DUE
Th-22jan09	25.5-25.6	
F-23 jan09	26.1-26.3	PS1 DUE
M-26 jan09	26.4-26.6	
W-28 jan09	26.7 - problems	
Th-29jan09	27.1-27.5	PS2 DUE
F-30jan09	27.6-27.7	
M-02feb09	problems, Guest Lecturer	
W-04feb09	**Exam1**: 14, 25, 26, 27	
Th-05feb09	28.1-28.4	
F-06feb09	28.4-28.8	
M-09feb09	28.9-problems	PS3 DUE
W-11feb09	29.1-29.4	
Th-12feb09	problems	
F-13feb09	30.1-30.3	PS4 DUE
M-16feb09	30.4-30.6	
W-18feb09	31.1-31.3	
Th-19feb09	31.4-31.6	
F-20feb09	31.7-problems	PS5 DUE
M-23feb09	32.1-32.3	
W-25feb09	32.4-32.6	
Th-26feb09	problems	PS6 DUE
F-27feb09	33.1-33.3	
M-02mar09	spring vacation	

T-03mar09	spring vacation	
W-04mar09	spring vacation	
Th-05mar09	spring vacation	
F-06mar09	spring vacation	
M-09 mar09	33.4-33.6	
W-11mar09	34.1-34.5	PS7 DUE
Th-12mar09	problems	
F-13mar09	35.1-35.4	
M-16mar09	APS problems, Guest Lecturer	
W-18mar09	APS problems, Guest Lecturer	
Th-19mar09	APS problems, Guest Lecturer	
F-20mar09	APS **Exam 2**, Ch 27 to 35	
M-23mar09	36.1-36.3	PS8 DUE
W-25mar09	36.4-36.5	
Th-26mar09	37.1-37.3	
F-27mar09	problems	PS9 DUE
M-30mar09	38.1-38.3	
W-01apr09	38.4-38.5	
Th-02apr09	38.7-problems	
F-03apr09	39.1-39.4	PS10 DUE
M-06apr09	39.5-39.6	
W-08apr09	40.1-40.4	
Th-09apr09	Holy Thursday-no classes	
F-10apr09	Good Friday-no classes	
M-13apr09	Easter Monday-no classes	
W-15apr09	40.5-40.6	
Th-16apr09	problems	PS11 DUE
F-17apr09	41.1-41.3	
M-20apr09	Patriot's Day	
W-22apr09	43.1- 43.2	PS12 DUE
Th-23apr09	43.3 - 43.4	
F-24apr09	problems	
M-27apr09	45.1-45.3	
W-29apr09	**Exam 3**, Ch 36-41, 43, 45	
Th-30apr09	review, problems	PS13 DUE
F-01may09	Study Day	
M-04may09	Study Day	
T-05may09	Term Exams	
W-06may09	Term Exams	
Th-07may09	Term Exams	
F-08may09	Term Exams	
S-09may09	Term Exams	
M-11may09	Term Exams	
T-12may09	Term Exams	
M-18 may09	Commencement	

Problem Set Schedule:

Code/Key:

MC = Multiple Choice Q = Questions E = Exercises P = Problems PS7= Problem Set Seven

Q: 28-4 (Chapter 28, Question Section, Question # 4) E: 25-6-29 (Chapter 25, Section 6, Exercise # 29)
P: 27-7 (Chapter 27, Problem #7)

PS0: Problems due for 14.1-14.7
MC: 14-2-1, 14-4-5, 14-6-8
Q: 14-3, 14-9, 14-38
E: 14-2-2, 14-4-5
P: 14-1, 14-4

PS1: Problems due for 25.1-25.6
Problem List:
MC: 25-2-3, 25-3-7, 25-4-10, 25-6-14
Q: 25-7, 25-11, 25-15, 25-28
E: 25-4-2, 25-4-6, 25-6-21, 25-6-29
P: 25-3, 25-4, 25-7, 25-10

PS2: Problems due for 26.1-26.7
Problem List:
MC: 26-2-1, 26-4-5
Q: 26-4, 26-5, 26-16
E: 26-2-1, 26-3-6, 26-5-22
P: 26-2, 26-5, 26-8, 26-10, 26-11

PS3: Problems due for 27.1-28.8
Problem List:
MC: 27-3-2, 27-4-4, 27-5-8, 28-4-3, 28-5-6
Q: 27-2, 27-7, 27-14, 28-3, 28-12, 28-17
E: 27-3-1, 27-4-7, 28-2-2, 28-3-6
P: 27-3, 27-5, 27-7, 27-8, 27-17, 28-1, 28-4, 28-10

PS4: Problems due for 28.9-29.4
Problem List:
MC: 28-8-16, 29.2-1, 29-3-6, 29-4-7, 29-5-11
Q: 28-30, 29-1, 29-7
E: 28-9-41, 29-3-4, 29-4-12
P: 29-1, 29-3, 29-7, 29-11, 29-14

PS5: Problems due for 30.1-31.6
Problem List:
MC: 30-2-2, 30-4-9, 30-5-11
Q: 30-4, 30-6, 31-8
E: 30-3, 30-5, 30-10, 30-14, 30-24, 31-2-3, 31-3-8, 31-5-20
P: 30-1, 30-4, 30-6, 30-9, 30-10, 31-6, 31-9

PS6: Problems due for 31.7-32.6
Problem List:
MC: 32-2-2, 32-2-4, 32-3-9
Q: 32-4, 32-17

E: 32-2-4, 32-2-6, 32-3-9, 32-3-14, 32-4-26, 32-6-36
P: 31-16, 31-17, 32-1, 32-3, 32-8, 32-11

PS7: Problems due for 33.1-33.6
Problem List:
MC: 33-1-1, 33-3-6
Q: 33-4
E: 33-2-4, 33-2-10, 33-2-16, 33-4-26
P:33-1, 33-2, 33-6, 33-9

PS8: Problems due for 34.1-34.4
Problem List:
MC: 34-2-1, 34-2-7, 34-5-18
Q: 34-1, 34-10
E: 34-2-1, 34-2-2, 34-3-3-4, 34-4-14
P: 34-1, 34-2, 34-9

PS9: Problems due for 34.5-36.5
Problem List:
MC: 34-5-13, 34-5-18, 36-2-2
Q: 34-14, 35-18, 36-5
E: 34-6-30, 35-1-10, 35-4-14, 35-4-18, 36-4-23
P: 35-1, 35-4, 35-8, 36-1, 36-4, 36-8

PS10: Problems due for 37.1-38.5
Problem List:
MC: 37-2-2, 38-1-1, 38-3-3, 38-5-7
Q: 37-1, 37-6, 38-10, 38-16
E: 37-3-14, 37-4-20, 38-2-7, 38-5-16, 38-5-17
P: 37-7, 37-11, 38-1, 38-3, 38-5, 38-10

PS11: Problems due for 39.1-40.5
Problem List:
MC: 39-3-2, 39-4-5, 39-5-11, 40-2-2, 40-5-10
Q: 39-6, 40-6, 40-18, 40-25, 40-34
E: 39-1-2, 39-1-4, 39-3-8, 39-4-14, 39-4-27, 40-2-2, 40-3-9, 40-3-15, 40-5-20
P: 39-4, 39-8, 39-12, 40-4, 40-8, 40-11

PS12: Problems due for 41.1-42.3
Problem List:
MC: 41-2-1, 41-2-5, 41-3-10
Q: 41-3, 41-19, 41-29
E: 41-2-1, 41-2-5, 41-3-10
P:41-1, 41-4, 41-6

PS13: Problems due for 43.1-43.4
Problem List:
MC: 43-1-1, 43-2-5
Q: 43-4, 43-13

E: 43-2-8, 43-3-16
P: 43-4

Grading of problem sets will follow the convention:
MC: 2/2 Q: 3/3 E: 5/5 P: 10/10 e.g. PS13: is worth 30/30 points.
Partial credit will be given for partial answers at the discretion of the graders. Appeals may be made to the instructor.